Part 1

Determine and explain with suitable examples (formula), Evaluation metrics for classification and clustering.

Solution:

For classification

1- Confusion Matrix

A table that is used to define the performance of a classification algorithm. Let's take example for binary classification (Yes/No). [20 Samples dataset]

Predicted class \ actual class	Yes	No
Yes	(TP) True positive	(FP) False Positive
No	(FN) False Negative	(TN) True Negative

Predicted class \ actual class	Yes	No
Yes	10	3
No	2	5

2- Accuracy

Proportion of correctly classified instances among the total.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

By using above (20 samples) dataset, lets calculate accuracy of binary (Yes/No) class.

$$\text{Accuracy} = \frac{10+5}{10+5+3+2} = \frac{15}{20} = 0.75 \cong 75\%$$

3- True positive

Instances correctly predicted as positive.

In above binary class (20 samples) of Yes/No.

$TP = 10$.

4- True Negative

Instances correctly predicted as negative.

$TN = 5$

Predicted class \ actual class	Yes	No
Yes	TP 10	3 FP
No	2 FN	TN 5

5- False positive

Instances incorrectly predicted as positive.

$FP = 3$

6- False Negative

Instances incorrectly predicted as negative.

$FN = 2$

7- True positive Rate (TPR) or Sensitivity or Recall

Proportion of actual positives correctly predicted.

$$TPR = \frac{TP}{TP + FN}$$

considering above example (Yes/No) class

$$TPR = \frac{10}{10+5} = \frac{10}{15} = 0.67$$

8- False Positive Rate (FPR)

Proportion of actual negatives incorrectly predicted as positive.

$$FPR = \frac{FP}{FP + TN}$$

considering above confusion matrix

$$FPR = \frac{3}{3+5} = \frac{3}{8} = 0.375$$

9- True Negative Rate (TNR) or Specificity.

Proportion of actual negatives correctly predicted.

$$TNR = \frac{TN}{TN + FP}$$

$$TNR = \frac{5}{5+3} = \frac{5}{8} = 0.63$$

10- False Negative Rate (FNR)

Proportion of actual positives incorrectly predicted as negatives.

$$FNR = \frac{FN}{TP + FN}$$

Predicted class		actual class Yes	actual class No
	Yes	10 TP	3 FP
	No	2 FN	5 TN

$$FNR = \frac{2}{10+2} = \frac{2}{12} = 0.17$$

11- Specificity

Proportion of actual negatives correctly predicted.

$$Specificity = \frac{TN}{TN + FP}$$

$$Specificity = \frac{5}{5+3} = \frac{5}{8} = 0.63$$

12- Sensitivity / Hit rate

Proportion of actual positives correctly predicted.

$$Sensitivity = \frac{TP}{TP + FN}$$

$$Sensitivity = \frac{10}{10+2} = \frac{10}{12} = 0.83$$

13- Precision

Proportion of predicted positives that are actually positive.

$$Precision = \frac{TP}{TP + FP} = \frac{10}{10+3} = \frac{10}{13} = 0.77$$

14- Recall

Proportion of actual positives correctly predicted.

$$Recall = \frac{TP}{TP+FN} = \frac{10}{10+2} = \frac{10}{12} = 0.83$$

15- F1 Score

Harmonic mean of precision and recall, balancing false positives and false negatives.

$$F1 = \frac{2 \times (Precision \times Recall)}{Precision + Recall} = \frac{2 \times (0.77 \times 0.83)}{(0.77 + 0.83)} = \frac{1.2782}{1.6} = 0.799$$

Part (II)

Categorization of classifiers based on metrics / performances:

a- Perfect classifier

characteristics: Achieves 100% accuracy, precision, recall and F1 score.

Metrics: Accuracy = 1.0, Precision = 1.0, Recall = 1.0, F1 Score = 1.0

b. Worst classifiers

characteristics: Performs very poorly, making the maximum number of errors.

Metrics: Accuracy = 0.0, Precision = 0.0, Recall = 0.0, F1 Score = 0.0.

c- Ultra liberal classifiers

characteristics: Labels almost everything as positive, minimizing false negatives but likely increasing false positives.

Metrics: High Recall, low precision

d- Ultra conservative classifies

characteristics: Labels very few instances as positive, minimizing false positives but likely increasing false negatives.

Metrics: High precision, low recall.

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(Part III)

For clustering:

- Internal measures

a- Cluster Cohesion

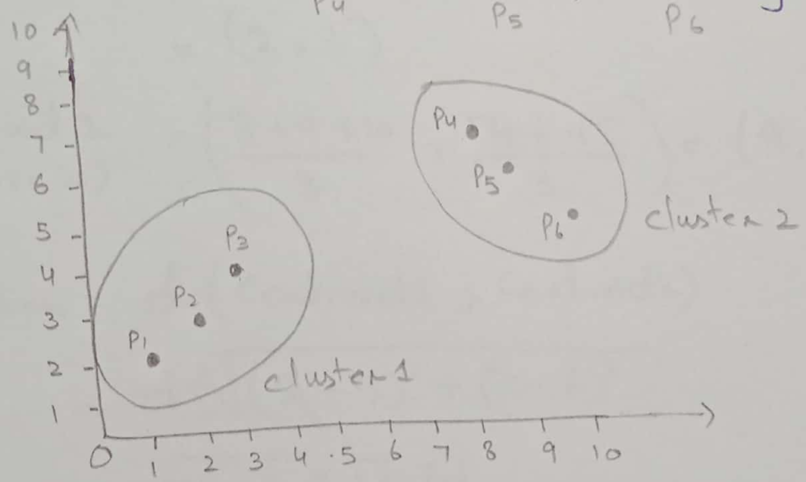
Measures how closely related the data points within a cluster.

Formula: Cohesion can be measured using the average Euclidean distance between points within same cluster.

Example

cluster 1 : $\{ \underset{P_1}{(1,2)}, \underset{P_2}{(2,3)}, \underset{P_3}{(3,4)} \}$

cluster 2 : $\{ \underset{P_4}{(8,7)}, \underset{P_5}{(9,6)}, \underset{P_6}{(10,5)} \}$



(By using Euclidean distance)

$$\text{Cohesion (cluster 1)} = \frac{d(1,2) + d(2,3) + d(3,4)}{3}$$

$$= \frac{\sqrt{(2-1)^2 + (3-2)^2} + \sqrt{(3-2)^2 + (4-3)^2}}{3}$$

$$= \frac{\sqrt{2} + \sqrt{2}}{3} = 0.94$$

$$\text{cohesion (cluster 2)} = \frac{\sqrt{(8-9)^2 + (7-6)^2} + \sqrt{(9-10)^2 + (6-5)^2}}{3}$$

$$= \frac{\sqrt{2} + \sqrt{2}}{3} = 0.94$$

Cohesion for Cluster 1 and 2 are equal. Since lesser the cohesion, better the clusters are.

b. Cluster Separation

Measures the separation between different clusters.

Formula: Separation can be measured using distance between cluster centroids.

Example

lets consider previous example of two clusters.

Cluster 1 = $\{(1,2),(2,3),(3,4)\}$

cluster 2 = $\{(8,7),(9,6),(10,5)\}$

Centroid 1 (cluster 1) $= \left(\frac{1+2+3}{3}, \frac{2+3+4}{3}\right)$

$= (2,3)$

Centroid 2 (cluster 2) $= \left(\frac{8+9+10}{3}, \frac{7+6+5}{3}\right) = (9,6)$

separation = d (Centroid1, Centroid2)

$= \sqrt{(2-9)^2+(3-6)^2}$

$= \sqrt{49+9} = 7.61$

Interpretation: Higher separation indicates better-defined clusters.

C- Silhouette Coefficients

Measures how similar an object is to its own cluster compared to other clusters.

Formula : $\text{Silhouette Coefficients} = \frac{b-a}{\max(a,b)}$

$\therefore$ a = average distance within a cluster.

b = average distance to the nearest neighbouring cluster.

Example : Let's take previous example of two clusters.

$\text{cluster } 1 = \{(1,2),(2,3),(3,4)\}$

$\text{cluster } 2 = \{(8,7),(9,6),(10,5)\}$

1- Compute a (average distance between/within cluster)

For cluster 1

$$\underset{(\text{cluster } 1)}{\text{Cohesion } 1} = \frac{d(1,2)+d(2,3)+d(3,4)}{3}$$

$$= \frac{\sqrt{2}+\sqrt{2}}{3} = 0.94$$

$$\underset{(\text{cluster } 2)}{\text{Cohesion } 2} = \frac{d(8,7)+d(9,6)+d(10,5)}{3}$$

$$= \frac{\sqrt{2}+\sqrt{2}}{3} = 0.94$$

2- Compute b (average distance to nearest neighbouring cluster)

For each point in cluster 1.

$b(1) = \min(d(1,8), d(1,9), d(1,10))$

$b(1) = \min(\sqrt{74}, \sqrt{85}, \sqrt{106})$

$b(1) = \sqrt{74}$

Similarly, we will compute b for each point in cluster 2.

Let's compute silhouette coefficient for each point in cluster 1:

$$\text{Silhouette coefficient}(1) = \frac{b(i) - \text{Cohesion1}}{\max(\text{Cohesion1}, b(i))}$$

$$= \frac{\sqrt{74} - 0.94}{\max(0.94, \sqrt{74})}$$

$$= \frac{\sqrt{74} - 0.94}{\sqrt{74}} = 0.789$$

Similarly, we will calculate the silhouette coefficients for other points in cluster 1, and repeat process for cluster 2.

Interpretation: A silhouette coefficient close to 1 indicates that the object is well matched to its own cluster and poorly matched to neighbouring clusters.

- External Measures

a- Entropy and purity

Measures the accuracy of a clustering algorithm by comparing it to a ground truth.

Entropy formula $= -\sum_{i=1}^{C} P(i) \sum_{j=1}^{K} P(j|i) \log_2 (P(j|i))$

$\therefore$ C = number of classes

K = number of clusters.

Purity formula $= \frac{1}{N} \sum_{K} \max_j |C_K \cap t_j|$

$\therefore$ N = total number of data points

Ck = Cluster K

t_j = ground truth class j.

Example let's take example of internal measures.

cluster1 = {(1,2), (2,3), (3,4)} ← True class 1

cluster2 = {(8,7), (9,6), (10,5)} ← True class 2

Entropy = $P(1|1) = 1$

For $i = 1$: $P(2|1) = 0$

$P(1|2) = 0$

$P(2|2) = 1$

Compute entropy for each i

$H(1) = -(1 \cdot \log_2(1) + 0 \cdot \log_2(0))$

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for $i = 2$

$$H(2) = -(0 \cdot \log_2(0) + 1 \cdot \log_2(1))$$

$$H = P(1) \cdot H(1) + P(2) \cdot H(2)$$

$$= \frac{3}{6} \cdot 0 + \frac{3}{6} \cdot 0 = 0$$

Entropy $= 0$

Interpretation: entropy (H) is 0, indicating perfect clustering with respect to true classes.

$$\text{Purity} = \frac{1}{N} \sum_k \max_j |c_k \cap t_j|$$

$$\text{Purity} = \frac{|\{(1,2),(2,3),(3,4)\} \cap \{(1,2),(2,3),(3,4)\}| + |\{(8,7),(9,6),(10,5)\} \cap \{(8,7),(9,6),(10,5)\}|}{6}$$

$$= \frac{3+3}{6} = \frac{6}{6} = 1$$

Interpretation: Purity is 1, indicating perfect clustering with respect to true classes.

b. Davies - Bouldin Index (DBI)

As we already computed Cohesion and separation values of following two clusters, let take this example.

Cluster 1 = {(1,2), (2,3), (3,4)}

cluster 2 = {(8,7), (9,6), (10,5)}

Cohesion 1 = 0.94

Cohesion 2 = 5.94

Centroid 1 = (2,3)

Centroid 2 = (9,6)

$$\text{Separation } 1,2 = \frac{0.94 + 5.94}{\sqrt{58}} = 0.903$$

Separation 2,1 = 0.903

Finally, Compute DBI

$$DBI = \frac{\text{Separation}_{1,2} + \text{Separation}_{2,1}}{2}$$

$$= \frac{0.903 + 0.903}{2} = 0.903$$

DBI = 0.903

Interpretation: lower DBI suggests that clustering has relatively good separation between clusters.

Proximity Matrix and Similarity Matrix

Proximity matrix is often a distance matrix, while similarity matrix is the inverse.

Proximity Matrix: (Euclidea distance between points)

Let's take data points for cluster 1 and cluster 2.

	(1,2)	(2,3)	(3,4)	(8,7)	(9,6)	(10,5)
(1,2)	0	$\sqrt{2}$	$2\sqrt{2}$	$\sqrt{61}$	$\sqrt{89}$	$\sqrt{117}$
(2,3)	$\sqrt{2}$	0	$\sqrt{2}$	$\sqrt{50}$	$\sqrt{65}$	$\sqrt{82}$
(3,4)	$2\sqrt{2}$	$\sqrt{2}$	0	$\sqrt{37}$	$\sqrt{50}$	$\sqrt{65}$
(8,7)	$\sqrt{61}$	$\sqrt{50}$	$\sqrt{37}$	0	$\sqrt{2}$	$2\sqrt{2}$
(9,6)	$\sqrt{89}$	$\sqrt{65}$	$\sqrt{50}$	$\sqrt{2}$	0	$\sqrt{2}$
(10,5)	$\sqrt{117}$	$\sqrt{82}$	$\sqrt{65}$	$2\sqrt{2}$	$\sqrt{2}$	0

Similarity Matrix: (using an inverse of distances)

	(1,2)	(2,3)	(3,4)	(8,7)	(9,6)	(10,5)
(1,2)	∞	$1/\sqrt{2}$	$1/2\sqrt{2}$	$1/\sqrt{61}$	$1/\sqrt{89}$	$1/\sqrt{117}$
(2,3)	$1/\sqrt{2}$	∞	$1/\sqrt{2}$	$1/\sqrt{50}$	$1/\sqrt{65}$	$1/\sqrt{82}$
(3,4)	$1/2\sqrt{2}$	$1/\sqrt{2}$	∞	$1/\sqrt{37}$	$1/\sqrt{50}$	$1/\sqrt{65}$
(8,7)	$1/\sqrt{61}$	$1/\sqrt{50}$	$1/\sqrt{37}$	∞	$1/\sqrt{2}$	$1/2\sqrt{2}$
(9,6)	$1/\sqrt{89}$	$1/\sqrt{65}$	$1/\sqrt{50}$	$1/\sqrt{2}$	∞	$1/\sqrt{2}$
(10,5)	$1/\sqrt{117}$	$1/\sqrt{82}$	$1/\sqrt{65}$	$1/2\sqrt{2}$	$1/\sqrt{2}$	∞

Part 3

Regression Evaluation Matrix

1- Mean Squared Error (MSE)

2- Root Mean Squared Error (RMSE)

3- Mean Absolute Error (MAE)

4- R squared

- Sum of Square total (SST)
- Sum of Squares regression (SSR)
- Sum of Squares Error (SSE)

Numeric Example:

Let's take following data for regression example.

Actual values: $y = [3, -0.5, 2, 7]$

predicted values: $\hat{y} = [2.5, 0.0, 2, 8]$

1- Mean Squared Error:

Measures the average of the squares of the errors.

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$MSE = \frac{1}{4} \left[(3-2.5)^2 + (-0.5-0.0)^2 + (2-2)^2 + (7-8)^2 \right]$$

$$= \frac{1}{4} [0.25 + 0.25 + 0 + 1] = 0.375$$

2- Root Mean Squared Error (RMSE)

square root of MSE, providing error in the same units as the target.

$$RMSE = \sqrt{MSE}$$

$$RMSE = \sqrt{0.375} = 0.612$$

3- Mean absolute Error (MAE)

Average of the absolute distance between predicted and actual values.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

using same y and $\hat{y}$ values as used previously,

$$MAE = \frac{1}{4}\left[|3-2.5| + |-0.5-0.0| + |2-2| + |7-8|\right]$$

$$= \frac{1}{4}\left[0.5 + 0.5 + 0 + 1\right] = 0.5$$

4- R-Squared (Coefficient of Determination)

Proportion of the variance in the independent variable that is predictable from the independent variables.

Let's calculate SSE and SST first.

- $SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

$$= \left[(3-2.5)^2 + (-0.5-0.0)^2 + (2-2)^2 + (7-8)^2\right]$$

$$= \left[(0.5)^2 + (-0.5)^2 + (0)^2 + (1)^2\right]$$

$$= 0.25 + 0.25 + 0 + 1 = 1.5$$

- $SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$ $\qquad \bar{y} = 2.625$

$$= (3-2.625)^2 + (-0.5-2.625)^2 + (2-2.625)^2 + (7-2.625)^2$$

$$= 20.75$$

- $SSR = R^2 = 1 - \frac{SSE}{SST}$

$$R^2 = 1 - \frac{1.5}{20.75} \simeq 0.928$$